

Python 101 Auto Grader

Design of the program

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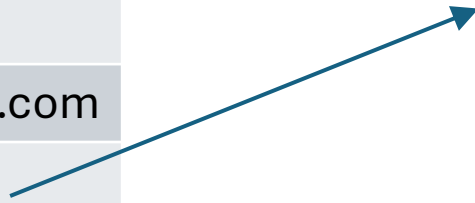
Design of the program

1. List of Students (where each item in the list reflects a student)
2. List of Assignments (where each item in the list reflects an assignment (name, due_date and max_points dictionary))
3. Dictionary of Emails (student_email is the key; And parents data is another nested dictionary)

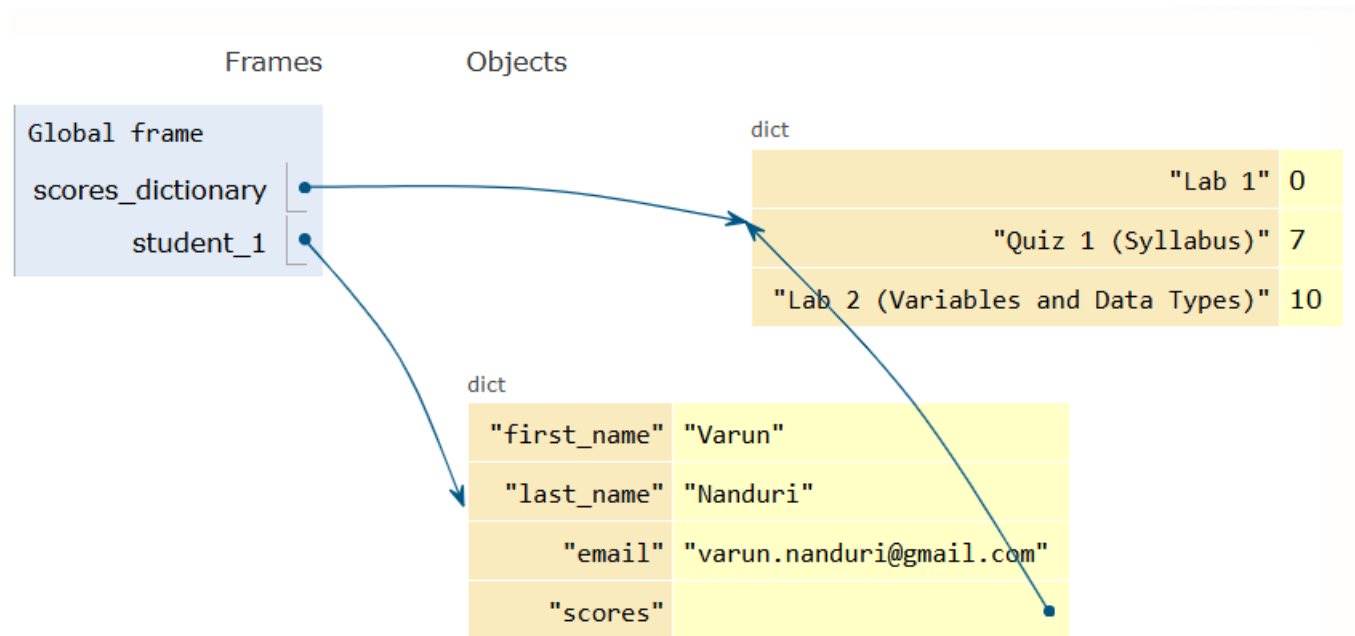
Representing a Student Data (dictionary containing a dictionary)

KEY	VALUE
last_name	ln1
first_name	fn1
email	fn1.ln1@learnandhelp.com
scores	

KEY	VALUE
Quiz1	None
Quiz2	None
Quiz3	10.0
lab1	10.0
	10.0
	...
	..
	...
	None
Final_project	None

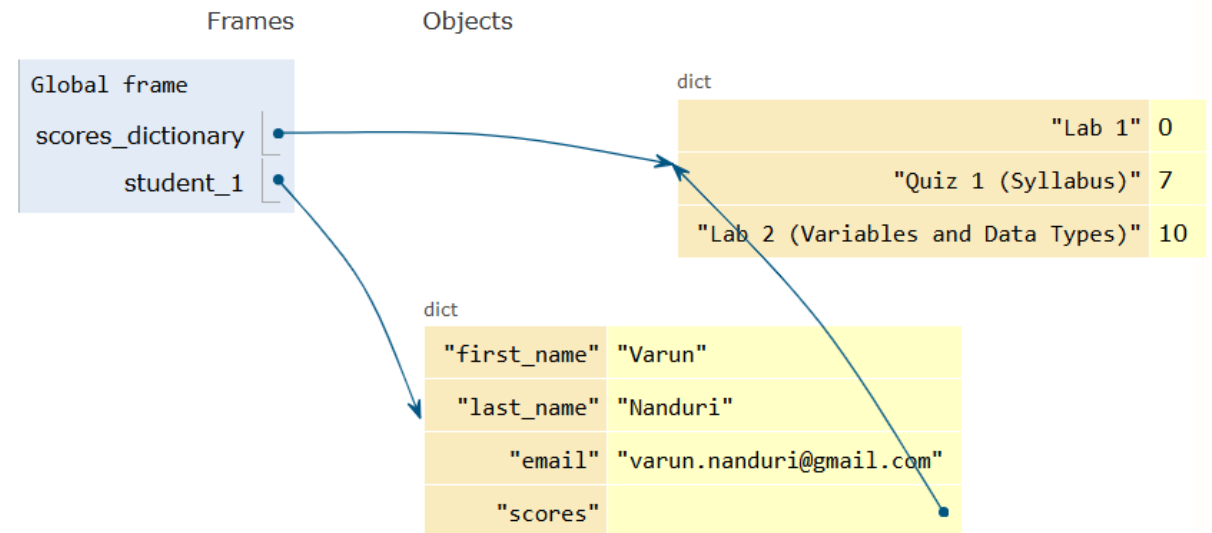


Representing a Student Data (dictionary containing a dictionary)



Representing a Student Data (dictionary containing a dictionary)

```
1
2 scores_dictionary = {"Lab 1": 0,
3                      "Quiz 1 (Syllabus)" : 7,
4                      "Lab 2 (Variables and Data Types)"
5                      }
6
7 student_1 = { "first_name" : "Varun",
8               "last_name" : "Nanduri",
9               "email" : "varun.nanduri@gmail.com",
10              "scores": scores_dictionary}
```



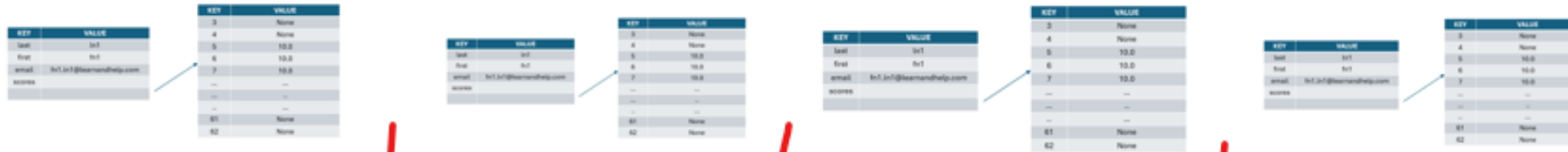
Representing two Students (dictionary containing a dictionary)

```
1
2 scores_1_dictionary = {"Lab 1": 0,
3                         "Quiz 1 (Syllabus)" : 7,
4                         "Lab 2 (Variables and Data Types)": 10
5                         }
6
7 student_1 = { "first_name" : "Varun",
8               "last_name" : "Nanduri",
9               "email" : "varun.nanduri@gmail.com",
10              "scores": scores_1_dictionary}
11
12
13 # student 2
14 scores_2_dictionary = {"Lab 1": 10,
15                         "Quiz 1 (Syllabus)" : 6,
16                         "Lab 2 (Variables and Data Types)": 10
17                         }
18
19 student_2 = { "first_name" : "Aditi",
20               "last_name" : "",
21               "email" : "aditi.boggarapu@gmail.com",
22               "scores": scores_2_dictionary}
23
24 # build the list
25 students_list = [student_1, student_2]
```

Representing all Students (list of student dictionaries)

student_data_with_scores (suggested variable name)

Python 101 Auto Grader
Design of the program



Python_101_Scores.csv

How to represent the assignments data?

While processing the scores, we can also populate the assignments list.

```
assignments_table = {"name": val,  
                    "max_points": val,  
                    "due_date": val}
```

KEY	VALUE
name	Lab 1
max_points	12
due_date	5/17/26

```
assignments_list = [assignments_table_1, assignments_table_2, and so on]
```

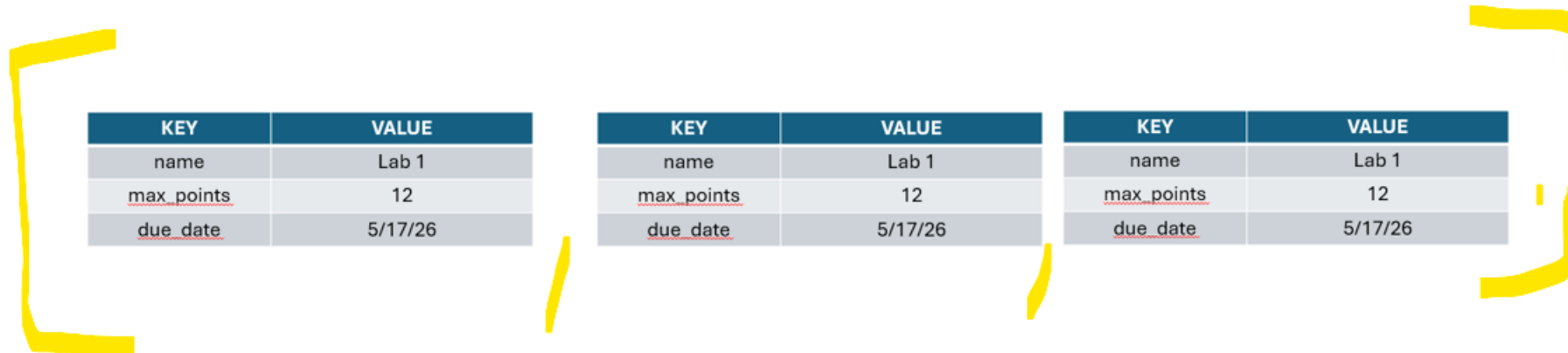

How to represent the assignments data?

While processing the scores, we can also populate the assignments list.

```
assignments_table = {"name": val,  
                    "max_points": val,  
                    "due_date": val}
```

KEY	VALUE
name	Lab 1
max_points	12
due_date	5/17/26

```
assignments_list = [assignments_table_1, assignments_table_2, and so on]
```



KEY	VALUE
name	Lab 1
<u>max_points</u>	12
<u>due_date</u>	5/17/26

KEY	VALUE
name	Lab 1
<u>max_points</u>	12
<u>due_date</u>	5/17/26

KEY	VALUE
name	Lab 1
<u>max_points</u>	12
<u>due_date</u>	5/17/26

How to represent email data?

How to load “python_101_emails.csv”?

Last Name

First Name

Email

Parent 1 Last Name,

Parent1 First Name,

Parent1 Email

Parent2 Last Name,

Parent2 First Name,

Parent2 Email

Dictionary of dictionaries

Where

KEY = students email

And VALUE = is another dictionary

KEY	VALUE
fn28.ln28@learnandhelp.com	



KEY	VALUE
p1_name	'P2FName ln28fn28 P1LName ln28fn28
p1_email	<u>P1_.fn28.ln28@learnandhelp.co</u> <u>m</u>
p2_name	'P2FName ln28fn28 P2LName ln28fn28
p2_email	'P2_.fn28.ln28@learnandhelp.c <u>om</u>

This is the sample data from the data structure

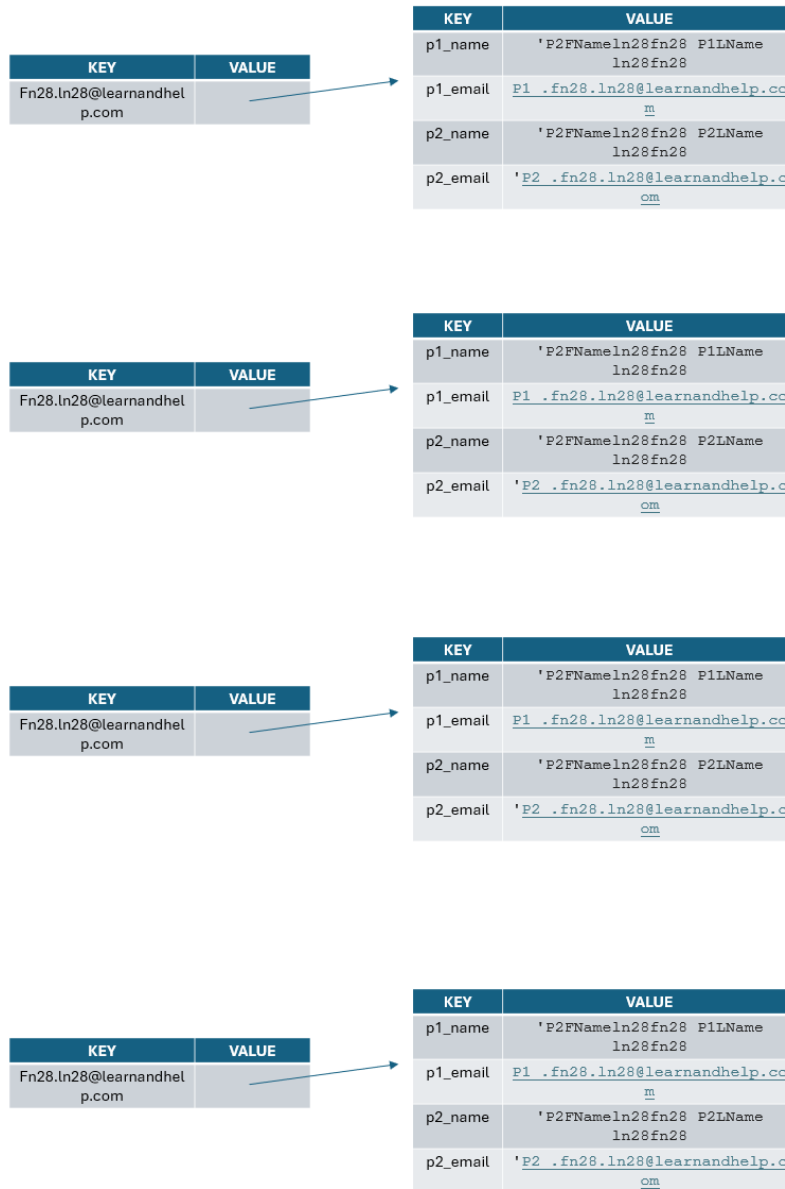
```
{ 'fn1.ln1@learnandhelp.com': {'p1_name': 'P2FNameIn1fn1 P1LName ln1fn1', 'p1_email':  
'P1 .fn1.ln1@learnandhelp.com', 'p2_name': 'P2FNameIn1fn1 P2LName ln1fn1', 'p2_email':  
'P2 .fn1.ln1@learnandhelp.com'}},  
  
'fn2.ln2@learnandhelp.com': {'p1_name': 'P2FNameIn2fn2 P1LName ln2fn2', 'p1_email':  
'P1 .fn2.ln2@learnandhelp.com', 'p2_name': 'P2FNameIn2fn2 P2LName ln2fn2', 'p2_email':  
'P2 .fn2.ln2@learnandhelp.com'}
```

We will have one dictionary.

Let us call it as "emails_lookup_table".

If we have 30 students, this "email_lookup_table" will have 30 keys.

Visual Representaton of emails_lookup_table



Plan of action

[1] Build "emails_lookup_table"

1. Load `python_101_emails.csv`
2. Read it and process it to build "emails_lookup_table"
3. Print it to ensure that you are getting all the details transferred from CSV to this data structure

[2] Build "students_data_with_scores"

1. Load `python_101_scores.csv`
2. Read it and process it to build "students_data_with_scores" (Please ensure that the first three rows are given special treatment - headers/labels, dates, and `max_permissible_points`)
3. Print it to ensure that you are getting all the details transferred from CSV to this data structure

Plan of action (contd.)

[2] Process both the data structures at the same time to build "a student's report" through a function.

[3] Process both the data structures at the same time to build "instructors" report (again through a function)

[4] Call all of the above functions to run the entire project.

- Read python_101_emails.csv and process it
- Read Python_101_scores.csv and process it
- Create Student Reports
- Create Instructor Report